

MUFAKOSE 1 HIGH MID-YEAR EXAMINATIONS

General Certificate of Education

Advanced Level

PHYSICS

Paper 3 ^{NOV}~~July~~ 2008

Time: 50mins

INSTRUCTIONS: Answer 3 questions

Question 1 is compulsory and any two remaining

Handwritten signature: Andrew Bellamy

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Answer question **number 1** and **any other 2** of the 3 remaining questions.

- 1 (a) In an experiment to determine the acceleration due to gravity g , the following results were obtained in ms^{-2} :

7.88; 7.86; 7.87; 7.86; 7.87.

- (i) Comment on the precision and accuracy of these results.
- (ii) Estimate the percentage uncertainty in the measurement of g .
- (iii) Suggest the type of error associated with the results.

[7]

- (b) (i) State the principle of superposition.
- (ii) Describe and explain how stationary waves can be produced in a stretched string.

[6]

- (c) Fig. 1.1 shows a body of mass 50 kg moving down a rough incline with a constant velocity v .

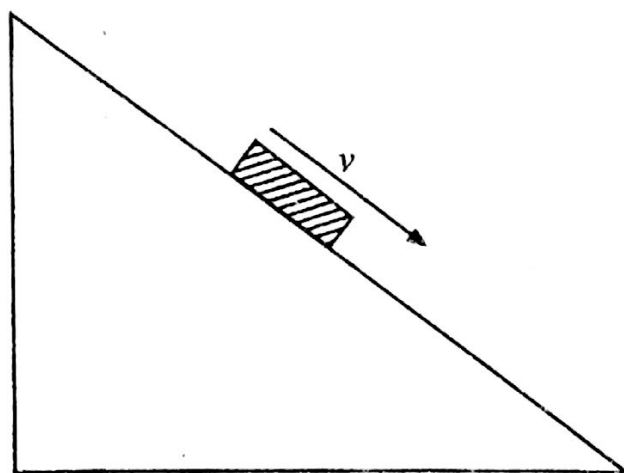


Fig. 1.1

- (i) Copy Fig. 1.1 and show all the forces acting on the body.
- (ii) Draw a triangle of forces for Fig. 1.1.
- (iii) Explain why the kinetic energy of the body remains constant despite the loss in gravitational potential energy.
- (iv) If the incline was smooth, what change will occur to the motion of the body?

[7]

- 2 (a) State **two** properties of electromagnetic waves. [1]
- (b) A and B in Fig. 2.1 are microwave transmitters, emitting waves that have a phase difference of π radians. O is the centre of a screen which is equidistant from A and B.

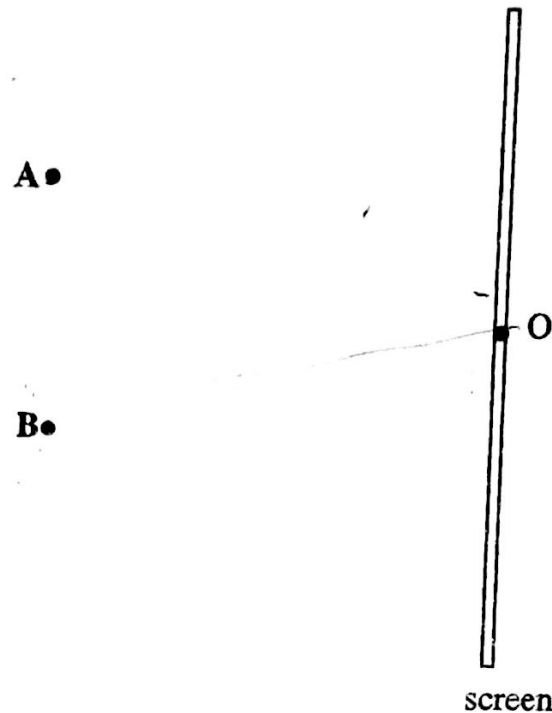


Fig. 2.1

- (i) Suggest a wavelength of microwaves and hence deduce a value for the separation of A and B.
- (ii) State and explain what is expected to be detected at O.
- (iii) Given that the screen is 2.50 m from sources A and B, use your values in (i) to calculate the separation of any two consecutive minima.

[7]

- (c) Explain whether it is feasible to make observations from your calculations in (b)(iii).

[2]

λ	1	λ	λ	λ	C
$10^{-2} - 10^{-4}$	10^{-4} $- 10^{-6}$	4.4×10^{-7} 7.8×10^{-7}	$10^{-2} - 10^{-1}$	10^{-2} 10^{-7}	$< 10^{-7}$

[Turn over

3 (a) State the conditions for a body to be in equilibrium. [2]

(b) A Physics teacher used a uniform metre rule to illustrate a point on the board. The rule was held at 32° to the horizontal as shown in Fig. 3.1.

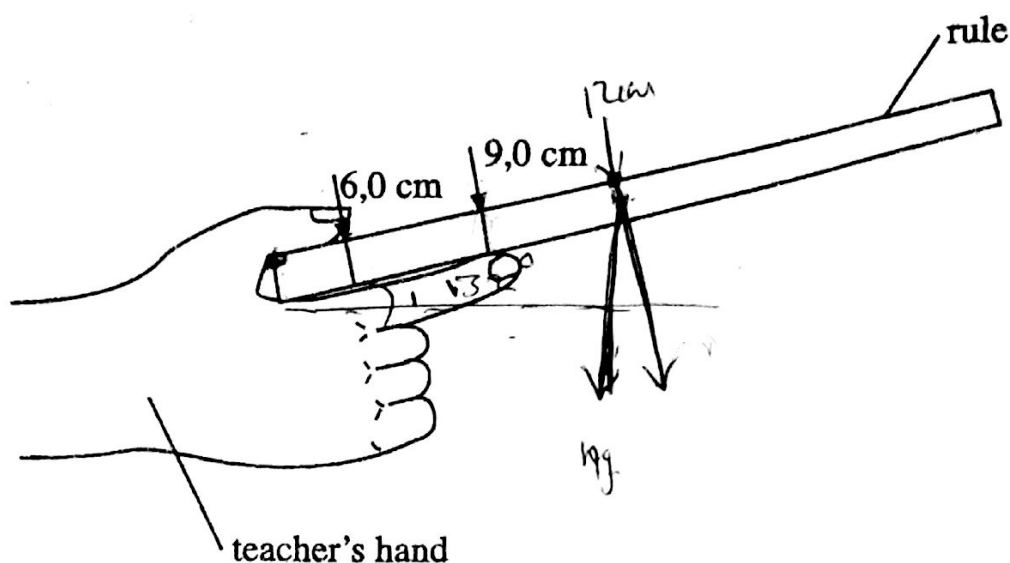


Fig. 3.1

(i) Make a sketch of Fig. 3.1 and draw all the forces acting on the rule.

(ii) Given that the mass of the metre rule is 107 g, determine the values of the forces that you have shown in (i). [6]

(c) State and explain an assumption that you have made in (b)(ii). [2]

4 (a) State Newton's second and third laws of motion. [2]

(b) A constant force of 9.5 N acts on a sphere A of mass 230 g lying on a smooth horizontal surface as shown in Fig. 4.1.

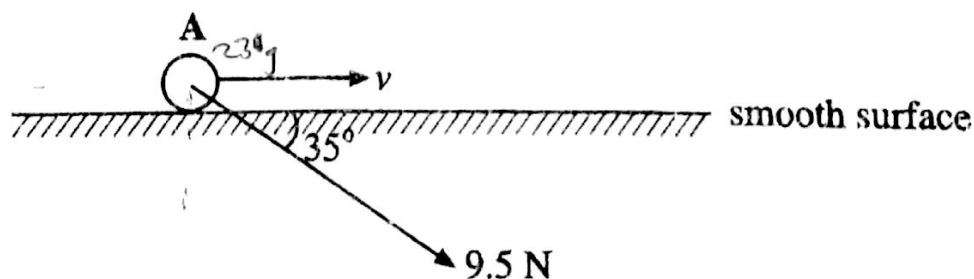


Fig. 4.1

$6.117(0)2 = 1$

The sphere A starts from rest and after 5.00 s the force is removed. Sphere A collides elastically with sphere B of mass 250 g initially at rest.

(i) Calculate

1. force exerted on sphere A by the surface,
2. the change in momentum of sphere A before collision,
3. the velocity of sphere A just before collision.

(ii) Write an equation relating the relative velocities in the collision. Hence or otherwise deduce the velocity of sphere A after the collision.

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